Personal Finance Dashboard – Final Project Report

Summary:
My project is a Personal Finance Dashboard built in Python. It allows users to upload a CSV file containing transaction data, and the program calculates the total income, expenses, and net balance. It also categorizes transactions, visualizes expenses by category with a bar chart, and provides budget warnings if spending exceeds set thresholds in specific areas like Food or Entertainment. A summary of the results is also exported to a text file for the user's records.

We used several key Python technologies to build this project:
- Object-oriented programming to model transactions and manage the dashboard
- The pandas library for data processing and CSV handling
- The matplotlib library for data visualization
- File I/O for reading input and exporting summary reports

Code Highlights:

1. Data Loading and Transformation:
We used `pandas.read_csv()` to load the transaction data from a file. The `FinanceManager` class parses each row into a `Transaction` object, making the data easier to work with.

2. Budget Warnings (Bonus Feature):
We implemented a custom function to sum category totals and display warnings if spending exceeds $200 on Food or $100 on Entertainment. This was useful in simulating budgeting scenarios and added meaningful functionality.

3. Visualization:
The `visualizer.py` file uses `matplotlib` to create a bar chart showing how expenses are distributed by category. This helps users identify where their money is going at a glance.

4. Exported Summary File:
We added a feature that writes a clear summary of income, expenses, and net balance to `summary.txt`. This makes the output usable beyond just the console and chart.

Screenshots of Important Code:
(Insert screenshots of key code sections, such as:
- Class definitions
- CSV reading logic
- Chart plotting function
- Budget warning logic)

More Ideas for a Future Version:
If I had more time, I would:
- Add support for monthly filtering or date-based summaries
- Create a full GUI using Streamlit or Tkinter for ease of use
- Integrate with live APIs like Plaid to pull transaction data from real bank accounts
- Allow users to set custom budget categories and thresholds

This project helped me reinforce Python skills, especially object-oriented design, data manipulation, and modular project structure. It's a tool we'd be proud to show friends or family who want to better understand their finances.

```python
transaction.py > Transaction > __init__
class Transaction:
    def __init__(self, date, description, category, amount):
        self.date = date
        self.description = description
        self.category = category
        self.amount = float(amount)
```

Figure 1: The Transaction class represents a single expense or income entry. Each instance stores the date, description, category, and amount.

```python
def main():
    manager = FinanceManager()
    manager.load_csv('transactions.csv')

    income, expense, balance = manager.get_summary()
    print(f"Total Income: ${income:.2f}")
    print(f"Total Expenses: ${abs(expense):.2f}")
    print(f"Net Balance: ${balance:.2f}")

    visualizer.plot_expense_by_category(manager.transactions)
    check_budget_warnings(manager.transactions)
    export_summary(income, expense, balance)
    print("Summary exported to summary.txt")

if __name__ == "__main__":
    main()
```

Figure 2: Where visualizer.plot begins, this part of the main() function calls our upgraded features: plotting a chart, warning if certain categories exceed a budget threshold, and exporting a report to summary.txt.

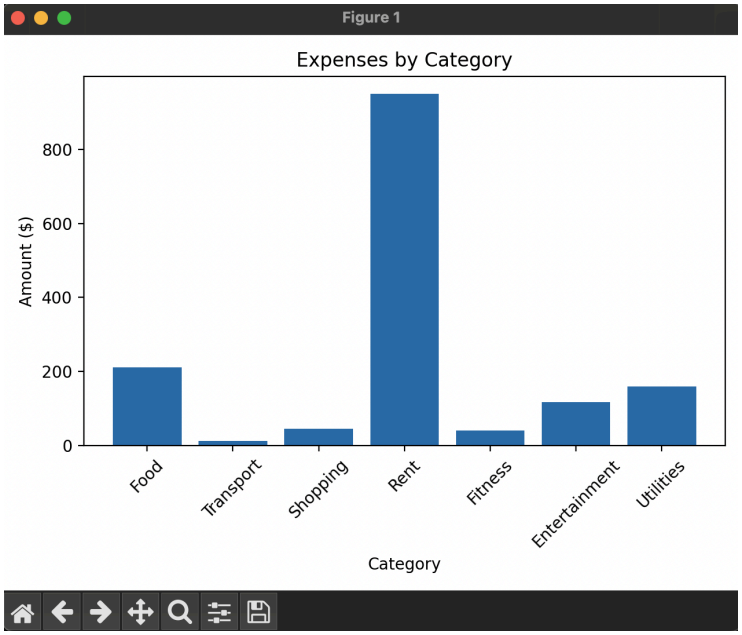

Figure 3: Bar chart visualization of expenses by category, generated using matplotlib.